

Antioxidants as potential therapeutics for neuropsychiatric disorders

Oxidative stress has been implicated in the pathophysiology of many neuropsychiatric disorders such as schizophrenia, bipolar disorder, major depression etc. Both genetic and nongenetic factors have been found to cause increased cellular levels of reactive oxygen species beyond the capacity of antioxidant defense mechanism in patients of psychiatric disorders. These factors trigger oxidative cellular damage to lipids, proteins and DNA, leading to abnormal neural growth and differentiation. Therefore, novel therapeutic strategies such as supplementation with antioxidants can be effective for long-term treatment management of neuropsychiatric disorders. The use of antioxidants and PUFAs as supplements in the treatment of neuropsychiatric disorders has provided some promising results. At the same time, one should be cautious with the use of antioxidants since excessive antioxidants could dangerously interfere with some of the protective functions of reactive oxygen species. The present article will give an overview of the potential strategies and outcomes of using antioxidants as therapeutics in psychiatric disorders.